# Filter Trait Outliers Using IQR

## Description

Removes outlier trait values from a long-format trait data frame using the interquartile range (IQR) method. Outlier detection is performed per group (by default per taxon x trait combination), mirroring the fence logic used by boxplots: \[Q1 - k*IQR, Q3 + k*IQR\].

## Usage

```
filter_trait_outliers(

  data_trait_records,

  trait_value_column = "trait_value",

  grouping_columns = c("taxon_name", "trait_domain_name"),

  iqr_multiplier = 1.5,

  verbose = TRUE

)
```

## Arguments

| | |
|---|---|
| `data_trait_records` | A data frame in long format with at least the columns specified by 'trait_value_column' and 'grouping_columns'. |
| `trait_value_column` | A character string naming the numeric column containing trait values. Default: '"trait_value"'. |
| `grouping_columns` | A character vector of column names used to define groups for outlier detection. Default: 'c("taxon_name", "trait_domain_name")'. |
| `iqr_multiplier` | A positive numeric scalar controlling the fence width, equivalent to k in the standard boxplot formula. Default: '1.5'. |
| `verbose` | Logical scalar. If 'TRUE', print the number of removed records. |

## Details

Groups with an IQR of zero (all values identical) are kept intact — no filtering is applied to constant groups. This prevents inadvertent removal of valid data when all observations share the same trait value.

## Value

A data frame with the same columns as the input but with outlier rows removed. The number of removed rows is reported via 'cli::cli_inform()'.

## See Also

[flag_trait_outliers()], [aggregate_trait_values()]